

1. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$\frac{27 + 22i}{8 - 7i}$$

- A. $62.00 + 3.23i$
 - B. $3.38 - 3.14i$
 - C. $0.55 + 365.00i$
 - D. $3.27 - 0.12i$
 - E. $0.55 + 3.23i$
-

2. Simplify the expression below into the form $a + bi$. Then, choose the intervals that a and b belong to.

$$(-9 - 5i)(10 - 4i)$$

- A. $-70 + 86i$
 - B. $-110 - 14i$
 - C. $-110 + 14i$
 - D. $-90 + 20i$
 - E. $-70 - 86i$
-

3. Simplify the expression below and choose the interval the simplification is contained within.

$$15 - 19^2 + 5 \div 3 * 20 \div 18$$

- A. -344.148
- B. *None of the above*
- C. 377.852
- D. 376.005

E. -345.995

4. Choose the **smallest** set of Real numbers that the number below belongs to.

$$-\sqrt{\frac{94864}{196}}$$

- A. Not a Real Number
 - B. Whole
 - C. Irrational
 - D. Rational
 - E. Integer
-

5. Choose the **smallest** set of Complex numbers that the number below belongs to.

$$\frac{-25}{23} + \sqrt{110}i$$

- A. Not a Complex Number
 - B. Nonreal Complex
 - C. Pure Imaginary
 - D. Irrational
 - E. Rational
-

6. Solve the equation below. Then, choose the interval that contains the solution.

$$-3(-4x - 8) = -2(16x + 18)$$

- A. -0.6
- B. -1.364
- C. There are no Real solutions.

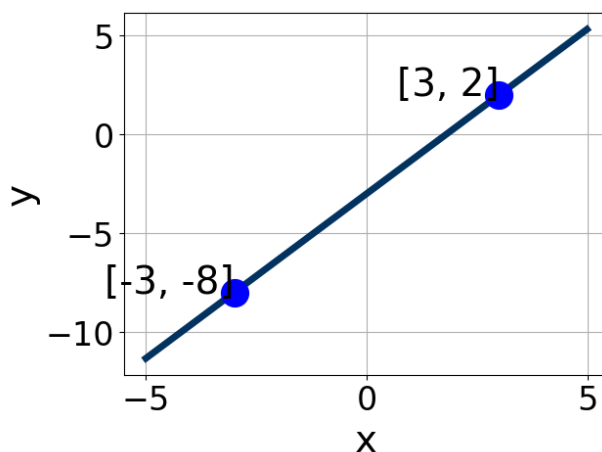
- D. 0.273
E. -0.273

7. Solve the linear equation below. Then, choose the interval that contains the solution.

$$\frac{7x - 4}{3} - \frac{7x - 6}{5} = \frac{8x + 5}{8}$$

- A. -47.375
B. There are no Real solutions.
C. -0.095
D. -45
E. -11.375

8. Write the equation of the line in the graph below in Standard Form $Ax + By = C$. Then, choose the intervals that contain A , B , and C .



- A. $-1.667x + 1y = -3.0$
B. $-5x + 3y = -9$
C. $5x - 3y = 9$
D. $5x + 3y = -9$
E. $-1.667x - 1y = 3.0$

-
9. First, find the equation of the line containing the two points below. Then, write the equation in the form $y = mx + b$ and choose the intervals that contain m and b .

$(-4, -8)$ and $(-10, 6)$

- A. $y = -2.33x + 17.33$
- B. $y = -2.33x + 16$
- C. $y = -2.33x - 4$
- D. $y = 2.33x + 29.33$
- E. $y = -2.33x - 17.33$

-
10. Find the equation of the line described below. Write the linear equation in the form $y = mx + b$ and choose the intervals that contain m and b .

Perpendicular to $7x - 4y = 7$ and passing through the point $(-7, -5)$.

- A. $y = 0.57x - 1.00$
- B. $y = -0.57x + 9.00$
- C. $y = -0.57x + 2.00$
- D. $y = -0.57x - 9.00$
- E. $y = -1.75x - 9.00$

-
11. Using an interval or intervals, choose the option that describes all the x -values within or including a distance of the given values.

Less than 3 units from the number 4.

- A. $(-\infty, 1) \cup (7, \infty)$
- B. $(-\infty, 1] \cup [7, \infty)$
- C. None of the above

D. $(1, 7)$

E. $[1, 7]$

-
12. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-7 + 4x \leq \frac{33x - 8}{7} < 8 - 4x$$

A. $(-\infty, 8.2) \cup [-1.049, \infty)$

B. $(8.2, -1.049]$

C. $(-\infty, 8.2] \cup (-1.049, \infty)$

D. $[8.2, -1.049)$

E. None of the above.

-
13. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-4 + 6x > 7x \text{ or } -4 + 6x < 8x$$

A. $(-\infty, -4.0)$ or $(-2.0, \infty)$

B. $(-\infty, 2.0]$ or $[4.0, \infty)$

C. $(-\infty, \infty)$

D. $(-\infty, -4.0]$ or $[-2.0, \infty)$

E. $(-\infty, 2.0)$ or $(4.0, \infty)$

-
14. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$-6x - 3 \leq 8x - 6$$

A. $(-\infty, -0.214]$

- B. $[-0.214, \infty)$
 - C. None of the above
 - D. $(-\infty, 0.214]$
 - E. $[0.214, \infty)$
-

15. Solve the linear inequality below. Then, choose the constant and interval combination that describes the solution set.

$$\frac{-8}{2} - \frac{6}{4}x > \frac{-5}{8}x + \frac{9}{6}$$

- A. None of the above
 - B. $(-\infty, 6.286)$
 - C. $(-\infty, -6.286)$
 - D. $(-6.286, \infty)$
 - E. $(6.286, \infty)$
-

16. Solve the quadratic equation below. Then, choose the intervals that the solutions x_1 and x_2 belong to, with $x_1 \leq x_2$.

$$10x^2 - 57x + 54$$

- A. $x_1 = 0.900$ and $x_2 = 6.000$
 - B. $x_1 = 1.200$ and $x_2 = 4.500$
 - C. $x_1 = 12.000$ and $x_2 = 45.000$
 - D. $x_1 = 0.400$ and $x_2 = 13.500$
 - E. $x_1 = 1.500$ and $x_2 = 3.600$
-

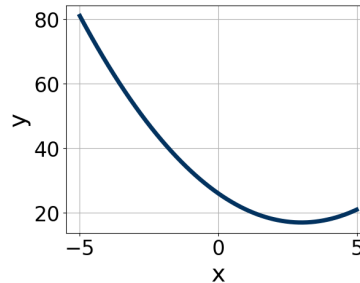
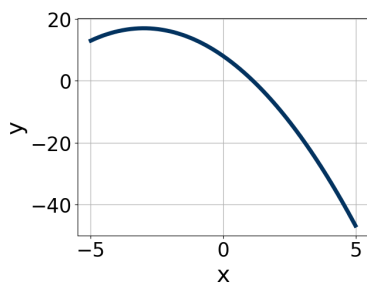
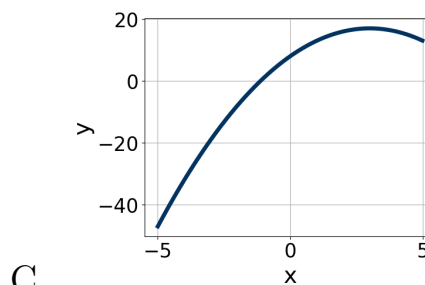
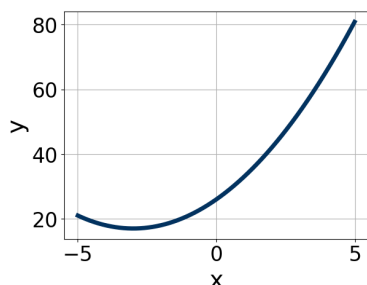
17. Factor the quadratic below. Then, choose the intervals that contain the constants in the form $(ax + b)(cx + d); b \leq d$.

$$90x^2 - 63x - 49$$

- A. $(12x - 7)(7x + 7)$
- B. $(6x - 7)(15x + 7)$
- C. $(2x - 7)(45x + 7)$
- D. $(x - 105)(x + 42)$
- E. None of the above.

18. Choose the graph of the equation below.

$$f(x) = -(x + 3)^2 + 17$$



- E. None of the above.

19. Solve the quadratic equation below. Then, choose the intervals that the solutions belong to, with $x_1 \leq x_2$ (if they exist).

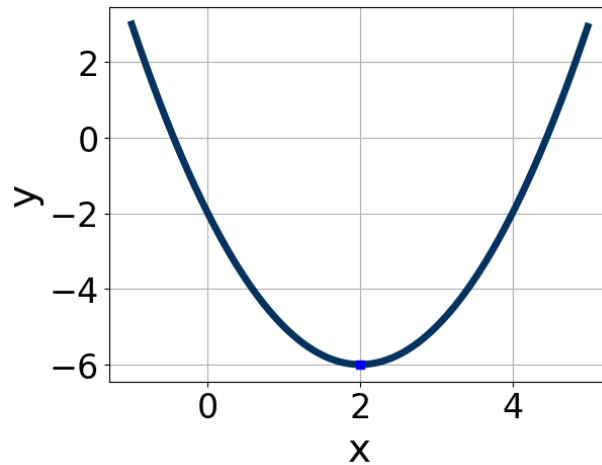
$$-10x^2 - 10x + 9$$

- A. $x_1 = 1.000$ and $x_2 = 9.000$
- B. $x_1 = -0.572$ and $x_2 = 1.572$
- C. $x_1 = -21.948$ and $x_2 = 20.948$

D. There are no Real solutions

E. $x_1 = -1.572$ and $x_2 = 0.572$

-
20. Write the equation of the graph presented below in the form $f(x) = ax^2 + bx + c$, assuming $a = 1$ or $a = -1$. Then, choose the intervals that a , b , and c belong to.



- A. $x^2 + 4x + 10$
B. $-x^2 + 4x - 10$
C. $x^2 + 4x - 2$
D. $-x^2 - 4x - 10$
E. $x^2 - 4x - 2$
-